$\mathcal{K}$	$\mathcal{K}^{\#1}_{0^+}$	$\mathcal{K}^{\#1}_{0^- \alpha\beta\chi}$
$\mathcal{K}^{\#1}_{0^+}\dagger$	0	0
$\mathcal{K}^{\#1}_{0^-}\dagger^{\alpha\beta\chi}$	0	$3\,k^2\,{}^{(4)}\kappa_{15} + 3\,{}^{(2)}\kappa_1$

$\mathcal{K}$	$\mathcal{K}^{\#1}_{1^+ \alpha\beta}$	$\mathcal{K}^{\#2}_{1^+ \alpha\beta}$	$\mathcal{K}^{\#1}_{1^- \alpha}$	$\mathcal{K}^{\#2}_{1^- \alpha}$
$\mathcal{K}^{\#1}_{1^+}\dagger^{\alpha\beta}$	$\frac{1}{2}\,(-\Upsilon_1\,k^2 + 4\,{}^{(2)}\kappa_1)$	$\frac{-\Upsilon_1\,k^2+4\,{}^{(2)}\kappa_1}{2\sqrt{2}}$	0	0
$\mathcal{K}^{\#2}_{1^+}\dagger^{\alpha\beta}$	$\frac{-\Upsilon_1\,k^2+4\,{}^{(2)}\kappa_1}{2\sqrt{2}}$	$\frac{1}{4}\,(-\Upsilon_1\,k^2 + 4\,{}^{(2)}\kappa_1)$	0	0
$\mathcal{K}^{\#1}_{1^-}\dagger^{\alpha}$	0	0	$-k^2\,{}^{(4)}\kappa_2$	$\frac{\Upsilon_2\,k^2}{2\sqrt{2}}$
$\mathcal{K}^{\#2}_{1^-}\dagger^{\alpha}$	0	0	$\frac{\Upsilon_2\,k^2}{2\sqrt{2}}$	$\frac{\Upsilon_3\,k^2}{4}$

$\mathcal{K}$	$\mathcal{K}^{\#1}_{2^+ \alpha\beta}$	$\mathcal{K}^{\#1}_{2^- \alpha\beta\chi}$
$\mathcal{K}^{\#1}_{2^+}\dagger^{\alpha\beta}$	$\frac{3\,\Upsilon_4\,k^2}{2}$	0
$\mathcal{K}^{\#1}_{2^-}\dagger^{\alpha\beta\chi}$	0	0

$\mathcal{J}$	$\mathcal{J}^{\#1}_{0^+}$	$\mathcal{J}^{\#1}_{0^- \alpha\beta\chi}$
$\mathcal{J}^{\#1}_{0^+}\dagger$	0	0
$\mathcal{J}^{\#1}_{0^-}\dagger^{\alpha\beta\chi}$	0	$\frac{1}{3\,(k^2\,{}^{(4)}\kappa_{15} + {}^{(2)}\kappa_1)}$

$\mathcal{J}$	$\mathcal{J}^{\#1}_{1^+ \alpha\beta}$	$\mathcal{J}^{\#2}_{1^+ \alpha\beta}$	$\mathcal{J}^{\#1}_{1^- \alpha}$	$\mathcal{J}^{\#2}_{1^- \alpha}$
$\mathcal{J}^{\#1}_{1^+}\dagger^{\alpha\beta}$	$\frac{2}{3\,\mathrm{Det}(1^+)}$	$\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}^{\#2}_{1^+}\dagger^{\alpha\beta}$	$\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)}$	$\frac{1}{3\,\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}^{\#1}_{1^-}\dagger^{\alpha}$	0	0	$\frac{\Upsilon_3\,k^2}{4\,\mathrm{Det}(1^-)}$	$-\frac{\Upsilon_2\,k^2}{2\sqrt{2}\,\mathrm{Det}(1^-)}$
$\mathcal{J}^{\#2}_{1^-}\dagger^{\alpha}$	0	0	$-\frac{\Upsilon_2\,k^2}{2\sqrt{2}\,\mathrm{Det}(1^-)}$	$-\frac{k^2\,{}^{(4)}\kappa_2}{\mathrm{Det}(1^-)}$

$\mathcal{J}$	$\mathcal{J}^{\#1}_{2^+ \alpha\beta}$	$\mathcal{J}^{\#1}_{2^- \alpha\beta\chi}$
$\mathcal{J}^{\#1}_{2^+}\dagger^{\alpha\beta}$	$\frac{1}{\mathrm{Det}(2^+)}$	0
$\mathcal{J}^{\#1}_{2^-}\dagger^{\alpha\beta\chi}$	0	0

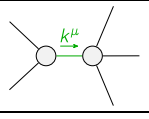
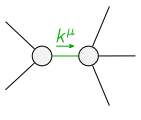
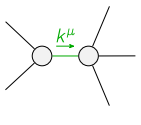
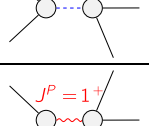
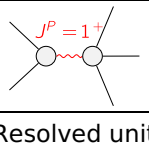
Abbreviations used in matrices

$$\Upsilon_1 == 3\,{}^{(4)}\kappa_1 - 4\,{}^{(4)}\kappa_{15} + 3\,{}^{(4)}\kappa_2 + 2\,{}^{(4)}\kappa_3 \;\&\&\; \Upsilon_2 == 2\,{}^{(4)}\kappa_2 + {}^{(4)}\kappa_7 \;\&\&\; \Upsilon_3 == 3\,{}^{(4)}\kappa_1 + {}^{(4)}\kappa_2 - 2\,{}^{(4)}\kappa_7 \;\&\&\; \Upsilon_4 == {}^{(4)}\kappa_1 + {}^{(4)}\kappa_2 \;\&\&$$
$$\mathrm{Det}(1^+) == -\frac{3}{4}\,k^2\,(3\,{}^{(4)}\kappa_1 - 4\,{}^{(4)}\kappa_{15} + 3\,{}^{(4)}\kappa_2 + 2\,{}^{(4)}\kappa_3) + 3\,{}^{(2)}\kappa_1 \;\&\&\; \mathrm{Det}(2^+) == \frac{3}{2}\,k^2\,({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)$$

Lagrangian

$$\begin{aligned}
&{}^{(2)}\kappa_1\,\mathcal{K}_{\alpha\beta\chi}\,\mathcal{K}^{\alpha\beta\chi} - 2\,{}^{(2)}\kappa_1\,\mathcal{K}^{\alpha\beta\chi}\,\mathcal{K}_{\beta\alpha\chi} + {}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} - \\
&3\,{}^{(4)}\kappa_1\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - 3\,{}^{(4)}\kappa_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - 2\,{}^{(4)}\kappa_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - 3\,{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \\
&3\,{}^{(4)}\kappa_2\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - {}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + {}^{(4)}\kappa_7\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \frac{3}{4}\,{}^{(4)}\kappa_1\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} - \\
&\frac{3}{4}\,{}^{(4)}\kappa_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} - \frac{1}{2}\,{}^{(4)}\kappa_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + {}^{(4)}\kappa_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} - 2\,{}^{(4)}\kappa_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}^{\#1\,\alpha\beta}_{1^+} == \mathcal{J}^{\#2\,\alpha\beta}_{1^+}$	3	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} + 2\,\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{2^-} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	9	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{9\,{}^{(4)}\kappa_1{}^2 + 9\,{}^{(4)}\kappa_2{}^2 + 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_7 + 4\,{}^{(4)}\kappa_7{}^2 + 6\,{}^{(4)}\kappa_1\,(3\,{}^{(4)}\kappa_2 + {}^{(4)}\kappa_7)}{({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)\,(6\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_2 + 6\,{}^{(4)}\kappa_2{}^2 + {}^{(4)}\kappa_7{}^2)}$
	1	0	$-\frac{1}{({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)\,(6\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_2 + 6\,{}^{(4)}\kappa_2{}^2 + {}^{(4)}\kappa_7{}^2)}\,(30\,{}^{(4)}\kappa_2{}^2 + 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_7 - {}^{(4)}\kappa_7{}^2 + 6\,{}^{(4)}\kappa_1\,(5\,{}^{(4)}\kappa_2 + {}^{(4)}\kappa_7) + \sqrt{(9\,{}^{(4)}\kappa_1{}^4 + 1269\,{}^{(4)}\kappa_2{}^4 - 24\,{}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_7 + 528\,{}^{(4)}\kappa_2{}^3\,{}^{(4)}\kappa_7 - 16\,{}^{(4)}\kappa_2{}^2\,{}^{(4)}\kappa_7{}^2 - 20\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_7{}^3 + 3\,{}^{(4)}\kappa_7{}^4 + 2\,{}^{(4)}\kappa_1{}^2\,(621\,{}^{(4)}\kappa_2{}^2 + 240\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_7 + 34\,{}^{(4)}\kappa_7{}^2) + 4\,{}^{(4)}\kappa_1\,(630\,{}^{(4)}\kappa_2{}^3 + 258\,{}^{(4)}\kappa_2{}^2\,{}^{(4)}\kappa_7 + 13\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_7{}^2 - 5\,{}^{(4)}\kappa_7{}^3)))}$
	1	0	$\frac{1}{({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)\,(6\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_2 + 6\,{}^{(4)}\kappa_2{}^2 + {}^{(4)}\kappa_7{}^2)}\,(-30\,{}^{(4)}\kappa_2{}^2 - 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_7 + {}^{(4)}\kappa_7{}^2 - 6\,{}^{(4)}\kappa_1\,(5\,{}^{(4)}\kappa_2 + {}^{(4)}\kappa_7) + \sqrt{(9\,{}^{(4)}\kappa_1{}^4 + 1269\,{}^{(4)}\kappa_2{}^4 - 24\,{}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_7 + 528\,{}^{(4)}\kappa_2{}^3\,{}^{(4)}\kappa_7 - 16\,{}^{(4)}\kappa_2{}^2\,{}^{(4)}\kappa_7{}^2 - 20\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_7{}^3 + 3\,{}^{(4)}\kappa_7{}^4 + 2\,{}^{(4)}\kappa_1{}^2\,(621\,{}^{(4)}\kappa_2{}^2 + 240\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_7 + 34\,{}^{(4)}\kappa_7{}^2) + 4\,{}^{(4)}\kappa_1\,(630\,{}^{(4)}\kappa_2{}^3 + 258\,{}^{(4)}\kappa_2{}^2\,{}^{(4)}\kappa_7 + 13\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_7{}^2 - 5\,{}^{(4)}\kappa_7{}^3)))}$
	1	$-\frac{{}^{(2)}\kappa_1}{{}^{(4)}\kappa_{15}}$	$-\frac{1}{3\,{}^{(4)}\kappa_{15}}$
	3	$\frac{4\,{}^{(2)}\kappa_1}{3\,{}^{(4)}\kappa_1 - 4\,{}^{(4)}\kappa_{15} + 3\,{}^{(4)}\kappa_2 + 2\,{}^{(4)}\kappa_3}$	$-\frac{4}{9\,{}^{(4)}\kappa_1 - 12\,{}^{(4)}\kappa_{15} + 9\,{}^{(4)}\kappa_2 + 6\,{}^{(4)}\kappa_3}$
Resolved unitarity condition(s):		(Demonstrably impossible)	